

Najmeh Nayyer

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RESEARCH INTERESTS

Combining experimental and theoretical approaches to elucidate neural principles underlying behavior and cognition, particularly perception and decision-making, with the eventual goal of applying this knowledge to AI.

EDUCATION

B.Sc., Psychology, University of Isfahan, Isfahan, Iran Sep 2024 - Present
GPA: 18.17/20 (3.89/4) Activities: Vice President of Psychology Students' Association
• Ranked 1135 (top 0.03%) among 348,917 participants in the national entrance exam for bachelors, which resulted in a full-ride scholarship.

SKILLS

Programming: Python and MATLAB
Libraries: NumPy, Pandas, SciPy, Matplotlib, Seaborn, PsychoPy, EEGLAB
Other Software: MS Office (PowerPoint, Word, Excel), SPSS
Research Skills: Statistical Analysis, Experimental Design, Data Visualization, Signal Processing (Training)
Soft Skills: Project Management, Event Management, Mentoring, Leadership, Self-Study
Languages: Persian (Native), English (Advanced)

EXPERIENCE

RESEARCH EXPERIENCE

Research Intern May 2026 - Present
Supervisor: Prof. Mohammadreza A. Dahaghani & Dr. Farideh Shakerian, IPM
Researcher Sep 2025 - Present
Supervisor: Prof. Maryam Vaziri-Pashkam, University of Delaware
• Behavioral study on object recognition using object kinematograms.
• Formed the study hypothesis, organized the literature review and analyzed the initial pilot data.
Project Assistant Apr 2025 - Aug 2025
Supervisor: Prof. Behrad Noudoost, University of Utah [GitHub Link](#)
• Intracocular transfer of color adaptation under varying ocular conditions.

MENTORING EXPERIENCE

Mentoring Intern, Cognitive Neuroscience Competition for High School Jan 2026 - Present
Iranian Neuroscience Society, Tehran, Tehran, Iran

PRESENTATIONS

Vahabi, F., Honarmand, F., **Nayyer, N.** (2025). *Chromatic Threshold Stability in The Yellow–Blue Axis*. Poster presented at the Basic and Clinical Neuroscience Congress, Tehran, Tehran, Iran ([Link](#))

SELECTED PROJECTS

Python Bootcamp Projects ([GitHub Link](#)) Ongoing
• Developed 10+ interactive Python projects including games (Snake, Pong, Turtle Crossing) and utilities (Coffee Machine inventory system, Flashcard app), demonstrating OOP, GUI design, and data management.
• Built functional applications spanning game logic, inventory simulation, and study tools using Python fundamentals including loops, conditionals, classes, and event-driven programming.
Analysis of Psychology Class of 2028 Social Status Dataset I Python ([GitHub Link](#)) Ongoing
• Used NumPy, Pandas, SciPy, Matplotlib, and Plotly library to preprocess and analyze an survey from our sociology class.
Memory-Guided Saccade Task I MATLAB ([Project Link](#)) June 2025
• Analyzed multi-unit spikes (25 neurons) and LFP from macaque FEF during a memory-guided saccade task. (8 angles, 2 radii) Plotted raster and PTSH.
• Decoded target location from population firing rates using kNN and characterized LFP power spectra using multitaper method.
• Computed mutual information and AUC to quantify neural encoding of angle and radius across task phases.